

Part A: Baseline

COL7524 – Advanced Computer Networks

Objective. The objective of this baseline experiment is to verify delivery of packets across the AS1 → AS2 → AS3 → AS4 path and compare the theoretical minimum one-way propagation delay against the average end-to-end delay reported by FlowMonitor.

1. Baseline Execution

Command used: ./ns3 run scratch/bgp

The simulation generated UDP traffic using the four-router baseline topology. The console output shows repeated client transmissions, server receptions, and replies, followed by successful client receptions. For instance, the first packet was sent at about +2.00000 s, received by the server at +2.01181 s, and the reply was received by the client at +2.02361 s. This confirms the end-to-end packet flow from AS1 to AS2 to AS3 to AS4 was successful.

2. Network Topology and Link Delays

The three point-to-point links in bgp.cc have the following propagation delays:

AS1	→	AS2	→	AS3	→	AS4
	3 ms		4 ms		3 ms	

3. Delay Calculation

The theoretical minimum one-way propagation delay is the sum of the delays of the three links:

$$D_{\text{theoretical}} = 3 \text{ ms} + 4 \text{ ms} + 3 \text{ ms} = 10 \text{ ms}$$

The FlowMonitor reports the following for the forward flow (flowId = 1) from interas-baseline-results.xml:

delaySum	$9.44533 \times 10^7 \text{ ns}$
rxPackets	8
lostPackets	0

$$\begin{aligned} \text{Average delay} &= \text{delaySum} / \text{rxPackets} \\ &= (9.44533 \times 10^7 \text{ ns}) / 8 \\ &= 1.18066625 \times 10^7 \text{ ns} \\ &\approx 11.807 \text{ ms} \end{aligned}$$

The reverse flow reports the same delaySum and rxPackets values, so it gives the same average delay.

4. Comparison and Explanation

Theoretical minimum propagation delay	10.000 ms
FlowMonitor average delay	11.807 ms
Difference	1.807 ms

The average delay of the FlowMonitor is about 1.807 ms above the theoretical minimum of 10 ms, accounting only for propagation. The 10 ms value is only for the configured propagation delay on the three links. FlowMonitor measures end-to-end packet delay, so the measured value can also include transmission/serialization and other processing or queuing effects in a simulated network. Hence, the difference indicates the difference between the propagation delay alone and the total packet delay as measured by FlowMonitor.

5. Conclusion

The baseline ns-3 simulation ran successfully and demonstrated packet flow along AS1 → AS2 → AS3 → AS4. The theoretical minimum one-way propagation delay was 10 ms, while FlowMonitor reported an average end-to-end delay of about 11.807 ms. The measured delay is slightly higher because it reflects the total end-to-end packet delay, not just the propagation delay.